
WORKSHOP #2 - THINKING



Presenter

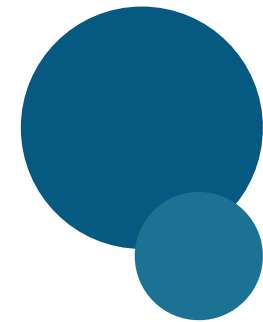
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Topic

AI Soup to Nuts

SESSION 2 OF 4 • TWO & HALF HOURS



Mental Models, Frameworks, and Judgment for a Fast-Moving Field

Building on Session 1's fundamentals — now we put that knowledge to work evaluating claims, spotting hype, and deciding when to trust AI.

Prerequisite: Session 1 — “AI Fundamentals” (or equivalent background knowledge)

WELCOME BACK

Quick recap: where Session 1 left off

History

AI has moved through hype cycles and winters since 1956 — today's moment isn't the first.

Vocabulary

Models, parameters, tokens, training, and inference — the building blocks of every AI conversation.

Mechanics

Neural networks learn through a repeated loop: predict, measure error, adjust weights.

Reality check

Today's AI is strong at pattern-based tasks, weak at verification and novel reasoning.

Today's goal: turn that background knowledge into judgment you can actually use.

A TWO-HOUR WORKSHOP

How to Think About AI

Building durable mental models for a fast-moving field — where AI came from, how it actually works, what it can't do, and how to evaluate the next big claim.

Duration: 2 hours · Format: Talk + discussion + exercise



AGENDA

What we'll cover in two hours 30 minutes

2:00 pm	Welcome & why this matters	Setting the goal: mental models, not just facts
2:10	Part 1 — Where AI came from	A fast history, from symbolic AI to deep learning
2:30	Part 2 — How AI actually works	The math and mechanics behind modern models
2:50	Exercises: First Look	10 minutes
3:00	Part 3 — What AI can and can't do	Capabilities, limitations, and common myths
3:25	Part 4 — Thinking critically about AI	A framework for evaluating claims and hype
3:45	Group exercise	Apply the framework to a real example
4:30	Wrap-up & resources	Key takeaways and where to go next (Workshop #3 & #4)

PART 1

Where AI Came From

A fast history — from ancient automata to Nobel-winning neural networks

PART 1 • HISTORY

A (very) condensed history of AI

1956

Dartmouth Conference coins the term “Artificial Intelligence”

1958

Rosenblatt's Perceptron — the first trainable neural network

1970s & 90s

“AI winters” as early hype outpaces results

1986

Backpropagation popularized by Rumelhart, Hinton & Williams

2006

Hinton's deep belief networks spark the deep learning resurgence

2012

AlexNet wins ImageNet — deep CNNs prove they scale

2017

“Attention Is All You Need” introduces the Transformer

2022+

ChatGPT & generative AI reach everyday use

2024

Hinton & Hopfield win the Nobel Prize in Physics

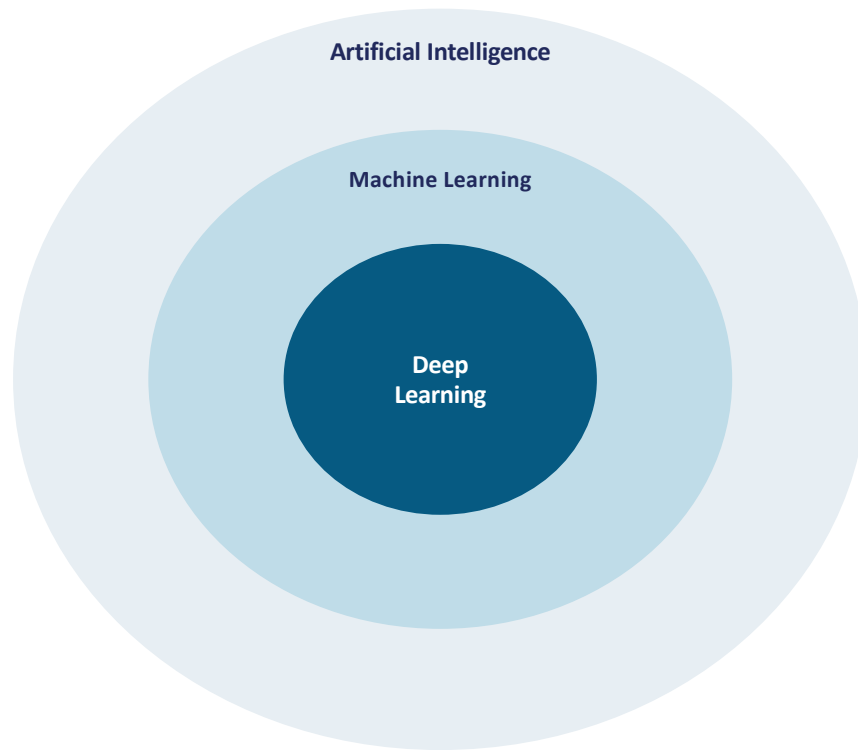
PART 2

How AI Actually Works

Peeling back the marketing language to see the mechanics underneath

THE BIG PICTURE

AI, machine learning, and deep learning are not the same thing



Artificial Intelligence

Any system designed to perform tasks that normally require human intelligence — including hand-coded rules from the 1960s.

Machine Learning

AI systems that improve by learning patterns from data, rather than following rules a person wrote by hand.

Deep Learning

Machine learning using many-layered neural networks — the approach behind image recognition, ChatGPT, and most recent AI progress.

UNDER THE HOOD

The math behind AI, in plain language



Linear Algebra

Data, weights, and activations are vectors and matrices. Every layer is fundamentally a matrix multiplication.



Calculus

Backpropagation uses derivatives (the chain rule) to work out how much each weight contributed to the error.



Probability & Statistics

Networks output distributions, not certainties. Loss functions measure how far predictions are from reality.

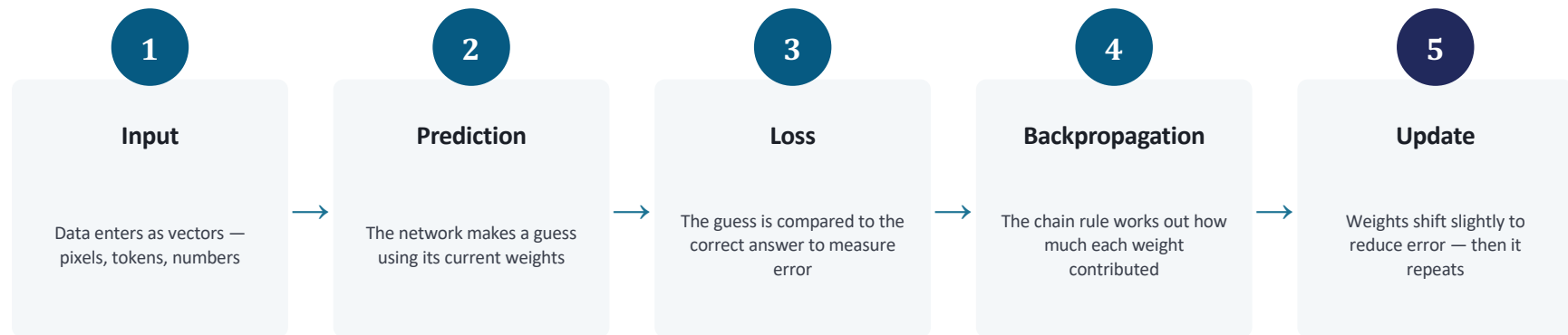


Optimization

Gradient descent repeatedly nudges weights toward lower error, gradually descending to a better model.

HOW TRAINING WORKS

The learning loop, one pass at a time



This loop repeats millions of times across a training dataset before a model is ready to use.

Pattern matching, not human reasoning

What it feels like

- Fluent, confident, human-like language
- Answers arrive instantly, fully formed
- Seems to “understand” context and intent

What's actually happening

- Predicting the statistically likely next token
- Drawing on patterns from training data, not lived experience
- No persistent beliefs, goals, or awareness between prompts

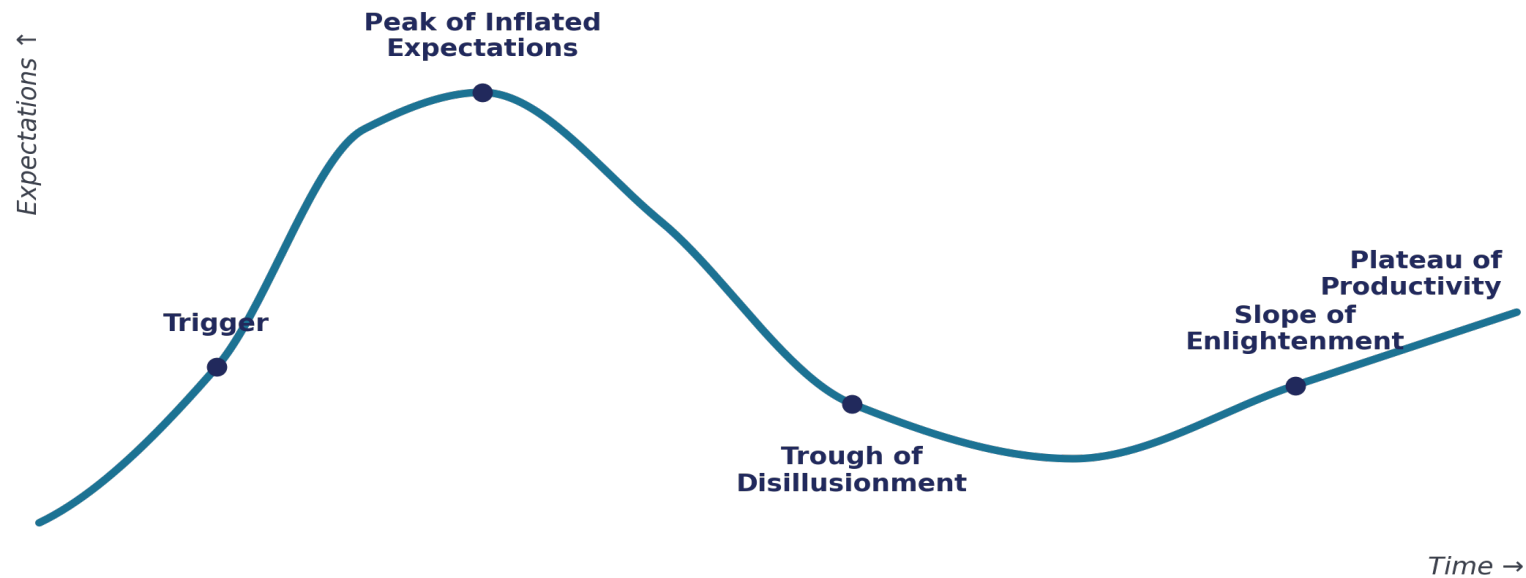
PART 3

What AI Can (and Can't) Do

Separating real capability from marketing language and hype

HYPE CYCLE

Every AI claim sits somewhere on this curve



Ask yourself: is this claim riding the peak of inflated expectations, or has it survived long enough to reach real productivity?

CAPABILITIES

Where AI is genuinely strong — and where it isn't



Strong at

- Summarizing and rephrasing large amounts of text
- Recognizing patterns across huge datasets
- Generating first drafts and boilerplate quickly
- Translating and transcribing language
- Writing and debugging routine code



Weak at

- Verifying its own claims are actually true
- Genuine novel reasoning outside its training patterns
- Maintaining consistent facts across a long session
- Understanding physical cause-and-effect it hasn't seen
- Knowing what it doesn't know

MYTHS

Common misconceptions worth retiring

“It understands what it's saying.”

It predicts plausible text based on patterns — with no inner model of truth or meaning.

“If it's confident, it's correct.”

Confidence and accuracy are unrelated. Hallucinated answers often sound the most fluent.

“More data always means a smarter model.”

Data quality, architecture, and training method matter as much as raw volume.

“It's basically a search engine.”

It generates new text from learned patterns — it doesn't look up or quote a source database.

PART 4

Thinking Critically About AI

A practical framework for evaluating claims, tools, and hype

FRAMEWORK

Four questions to ask about any AI claim

1

What is it actually trained or built to do?

Separate the underlying mechanism from the marketing description.

2

What's the evidence, and who is it from?

Independent benchmarks and peer review carry more weight than a product launch post.

3

What happens when it's wrong?

Understand the failure mode and its real-world cost before you rely on it.

4

Who benefits from you believing this claim?

Vendors, researchers, and critics all have different incentives — read accordingly.

WORKED EXAMPLE

Applying the framework to a real-style claim

"The Claim"

"A new model achieves expert-level performance on a professional licensing exam."

1

What's it built to do?

Trained to predict likely answers to exam-style questions — not to practice the profession itself.

2

What's the evidence?

Check whether the score comes from the vendor's own test or an independent, replicated evaluation.

3

What happens when wrong?

An exam question has one right answer; real-world cases are messier and the cost of an error is much higher.

4

Who benefits?

A strong benchmark result is a powerful marketing headline for whoever released the model.

GROUP EXERCISE • 15 MINUTES

Apply the framework

In pairs, pick a recent AI headline or product claim you've seen this month. Walk it through all four questions from the framework, then be ready to share:

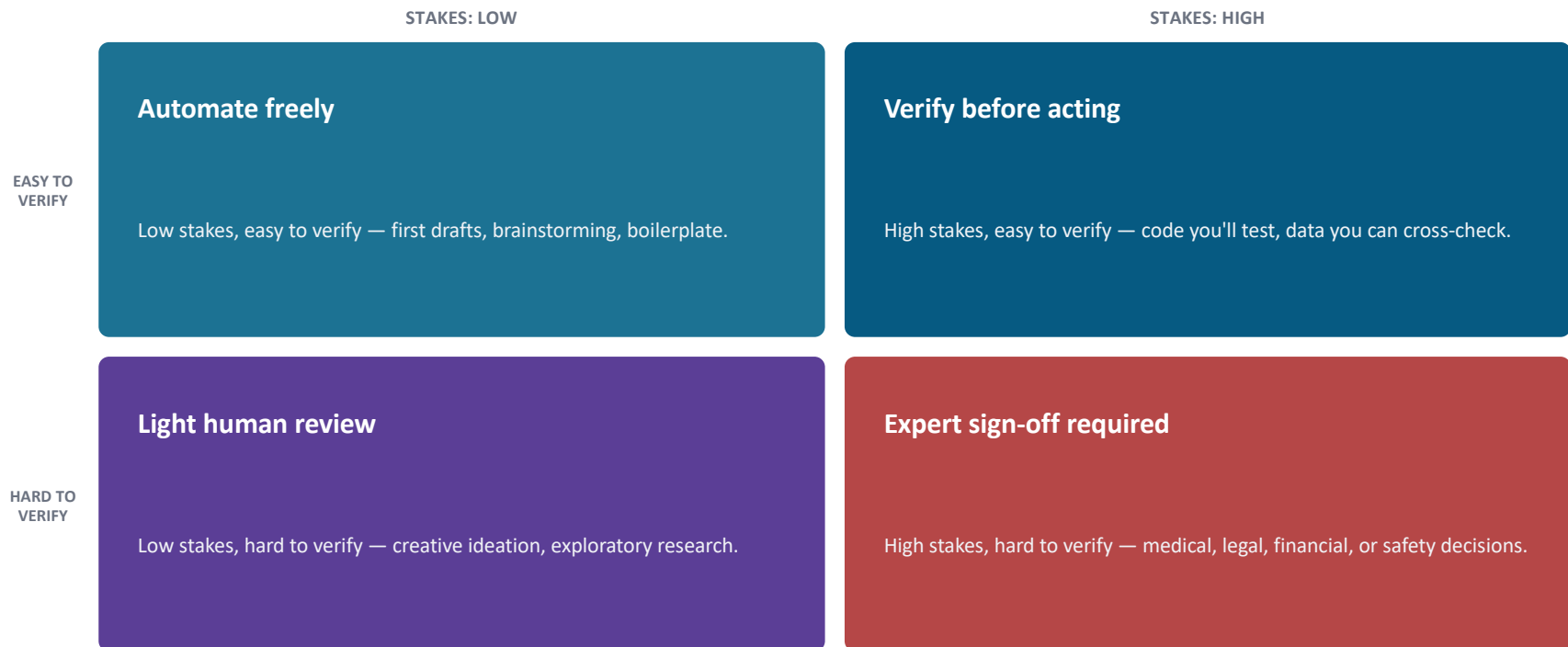
What did the claim get right, if anything?

What would you want to see before believing it?

How would you explain it to someone non-technical?

DECISION FRAMEWORK

Match the tool to the stakes and the verifiability



Wrap-Up

Key mental models to take with you

TAKEAWAYS

Five mental models worth keeping

1

History repeats

AI has gone through hype cycles and winters before — enthusiasm and skepticism should both be calibrated.

2

It's pattern matching

Modern AI predicts likely patterns from data; it doesn't reason, believe, or understand the way people do.

3

Confidence ≠ correctness

Fluent, assured answers can still be completely wrong. Always verify anything that matters.

4

Ask who benefits

Every AI claim comes from somewhere. Consider the incentive behind the message before adopting it.

5

Match the tool to the task

AI is a strong assistant for drafting and pattern-finding — not a substitute for judgment on high-stakes decisions.

Thank you

Questions, discussion, and further reading

Hinton, Osindero & Teh (2006)

“A Fast Learning Algorithm for Deep Belief Nets” — Neural Computation

Vaswani et al. (2017)

“Attention Is All You Need” — introduced the Transformer architecture

Krizhevsky, Sutskever & Hinton (2012)

The AlexNet paper — the ImageNet result that restarted the field

The Nobel Prize in Physics 2024

Official citation for Hinton & Hopfield's award